Motor Imagery Movements Classification Using Multivariate EMD and Short Time Fourier Transform

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***Abstract*—In this paper, a novel method consisting of multivariate empirical mode decomposition (MEMD) and short time Fourier transform (STFT) is proposed to identify left and right imagery hand movements from electroencephalogram (EEG) signals. Experiments are carried out using the publicly available benchmark BCI competition 2003 Graz motor imagery data base of left and right hand movements. The EEG epochs are decomposed into multiple intrinsic mode functions (IMFs) by applying MEMD. The most significant mode is subjected to the short time Fourier transform; the peak and entropy of the magnitude spectrum are used as features representing the corresponding epoch. Extensive analysis is carried out using Kruskal-Wallis test, scatter plots, box plots and histograms to justify the employed features. Classification of the motor imagery movements is studied using the proposed features and various state of the art classifiers. The highest accuracy is achieved employing the k-nearest neighbor (kNN) classifier which is 90.00% and better than those of the several contemporary methods.**

Keywords*—Electroencephalogram (EEG); Multivariate EMD; Entropy; Short time Fourier transform; kNN classifier.

I. Introduction

Brain computer interfacing (BCI) allows to control and operate computer aided systems by intent alone. BCI involves detection, analysis and classification of different types of motor imagery movements to implement real time control and communication. The major objective of BCI is to assist disable people for their rehabilitation. Electroencephalogram (EEG) signals are often used for BCI purpose since it can be implemented as a non-invasive system [1].

There are several categories of EEG-based BCI such as limb motor imagery classification [2], continuous arm movements direction detection [3] and individual finger movement decoding [4] etc. One major category of BCI is the detection of motor imagery movements such as left and right hand movements [5]. Various methods have been developed in the literature for classifying different types of arm movements. A wavelet-based common spatial pattern (CSP) algorithm using low frequency features and Fisher linear discriminant classifier is developed to classify fast and slow hand movements in [6]. In [7], filter bank common spatial pattern (FBCSP) is implemented with mutual information based feature selection and for the identification task, Naive Bayesian Parzen Window (NBPW) classifier is used. Wrist movement classification has been done by extracting gamma band features from wavelet packet transform and employing radial basis function (RBF) classifier in [8].

Since EEG data for research purpose can be acquired with varying experimental setup and conditions (e.g., [9]), BCI competition was held providing standard data sets to evaluate and compare different algorithms. The standard data sets were proved to be representative in motor imagery and were suitable for BCI research. Different approaches have been studied to classify motor imagery movements in BCI competition 2003 Graz motor imagery data set. Band passed EEG signals and power spectral density based linear discriminant analysis (LDA) is proposed in [10]. A Hidden Markov Model (HMM) based method is presented in [11] by the same author. Adaptive Auto Regressive (AAR) model based features are used with Bayesian Graphical Network (BGN) and Multi Layer Perceptron classifiers in [12]. Morlet wavelet is used to extract features from mu rhythms that are used in Bayes quadratic classifiers in [13].

The objective of this work is to identify imagery hand movements by extracting suitable features from the EEG signals using the multivariate EMD (MEMD) and STFT based hybrid approach. Since EEG signal is invariably nonlinear and non-stationary [14], fixed linear orthogonal basis functions are not suitable for real life EEG data. The underlying dynamics of EEG signals is spread over various sub-bands in the frequency domain and particularly for motor imagery analysis, mu (8-12 Hz) and beta (18-25 Hz) rhythms have significant importance in neurophysiological context [15]. Empirical mode decomposition (EMD) has been successfully utilized in the processing of EEG signals [16]- [17]. However, EEG signals are usually multi-channel type whereas the EMD is applied on a single-channel basis. Recently, the multivariate EMD (MEMD) has been introduced which can capture the cross-channel dependency and can be applied directly to all the EEG channels. Thus, in this paper the multi-channel EEG signal is decomposed into various intrinsic mode functions (IMFs) using MEMD. Furthermore, in order to obtain localized information, the most significant intrinsic mode function is subjected to STFT and the peaks along with entropy of the magnitude spectrum are used as features with sufficient justifications.

II. Description of the EEG database

BCI competition 2003 data set (GRAZ motor imagery III) provided by Technical University of Graz is used in this paper. The data is acquired from a normal subject with the subject sitting in a chair with armrests and trying to control a feedback bar by making imagery movements of left or right hands. Left and right cues are in random order [18]. During each trial at

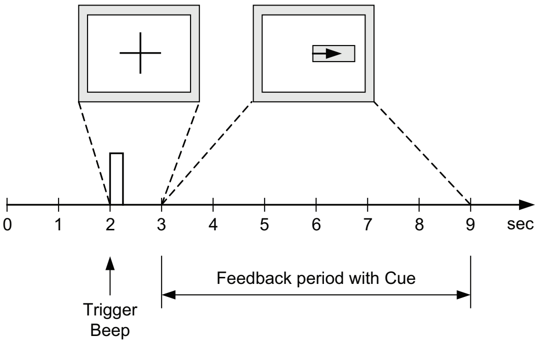

Fig. 1: Timing scheme of the experiment.

2s, an acoustic stimulus indicates the beginning of the trial and a cross '+' is displayed for 1s. After this an arrow (left or right) is displayed at 3s as the cue. At the same time the subject is asked to move a bar into the direction of the cue which is controlled by adaptive auto-regressive parameters of channel C3 and C4. The EEG signal is filtered between 0.5 and 30 Hz while the sampling rate is 128 Hz. Fig. 1 shows the timing scheme of the EEG signal acquisition system. A detail description of the data set can be found in [19].

III. Multivariate Empirical Mode Decomposition

The multivariate empirical mode decomposition (MEMD) is recently developed, where instead of computing the local mean using the average of upper and lower envelopes like conventional EMD, the multiple n dimensional envelopes are generated by projecting the signal along every directions in n variate spaces in MEMD [20]. These projections are averaged to obtain the local mean.

Let a multivariate signal with n components is denoted by n dimensional vectors $\{\mathbf{v}(t)\}_{t=1}^{T} = \{v_1(t), v_2(t), \cdots, v_n(t)\}$ where $\mathbf{x}^{\theta_k} = x_1^k, x_2^k, \cdots, x_n^k$ denotes a set of direction vectors along the directions given by angles $\theta^k = \{\theta_1^k, \theta_2^k, \cdots, \theta_{(n-1)}^k\}$ on an $(n-1)$ sphere. The steps to compute MEMD is given below [21]:

1. Select suitable points for sampling on an $(n-1)$ sphere
2. Calculate projection $\{p^{\theta_k(t)}\}_{t=1}^{T}$ along the direction vector $\mathbf{x}^{\theta_k}$ of the input signal $\{\mathbf{v}(t)\}_{t=1}^{T}$ for all k resulting $\{p^{\theta_k(t)}\}_{k=1}^{K}$ as the projection set.
3. Find the time instants $\{t_i^{\theta_k}\}$ corresponding to the maxima of $\{p^{\theta_k(t)}\}_{k=1}^{K}$.
4. Interpolate $[t_i^{\theta_k}, \mathbf{v}(t_i^{\theta_k})]$ to obtain multivariate envelope curves $\{\mathbf{e}^{\theta_k}(t)\}_{k=1}^{K}$.
5. Calculate the mean $\mathbf{m}(t)$ of the envelopes for K direction vectors as

$$\mathbf{m}(t) = \frac{1}{K}\sum_{k=1}^{K}\mathbf{e}^{\theta_k}(t) \qquad (1)$$

6. Extract the detail $d(t)$ using $d(t) = x(t) - m(t)$. If $d(t)$ fulfills the stoppage criterion for the IMF, apply the procedure to $x(t) - d(t)$, else apply it to $d(t)$.

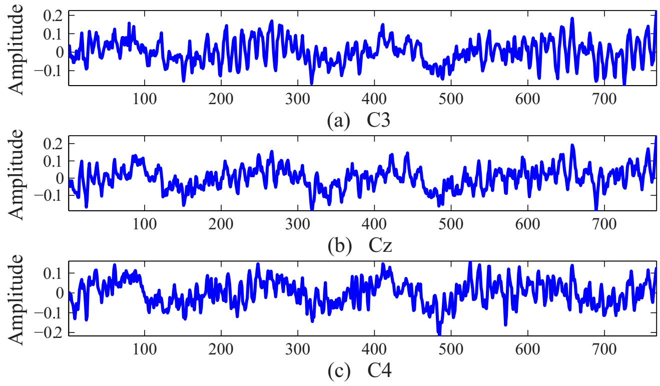

Fig. 2: Original EEG signals: collected from (a) C3 channel, (b) Cz channel and (c) C4 channel, respectively.

Fig. 2 shows the original EEG signals collected from C3, Cz and C4 channels, respectively. In Fig. 3, 8 IMFs for C3 channel generated from MEMD are presented.

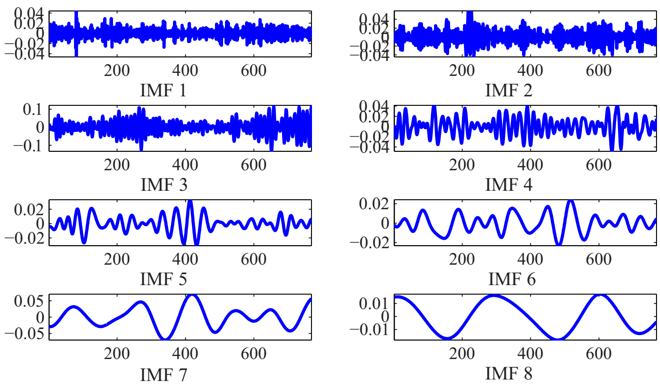

Fig. 3: 8 IMFs of EEG signal for C3 channel.

IV. Analysis in Multivariate EMD Domain

The multivariate EMD of the EEG signal leads to multiple intrinsic mode functions (IMFs). In this study eight IMFs have been extracted from EEG signal. However, all the IMFs are not equally significant in the detection of left and right hand imagery movements. To select the most significant IMF, we have analyzed the energy of different IMFs. Fig. 4 (a) shows that the energy of the third IMF is significantly greater than any other IMF for left and right hand movements and as a result, it is selected for further analysis.

A. Feature Extraction from the IMFs

The activation of brain signals for left and right hand differs in time frequency scale in the IMF and in order to separate two different hand imagery signals by extracting suitable features, we need to look for the "activated" frequency components of these signals [22]. For analyzing the activated frequency components, power spectral density (PSD) is obtained for these two signals. One can see from Fig. 5 that PSD of left and right hand imagery movements are quite similar. It indicates that these signals do not have distinguishing features in the Fourier domain. To extract effective features for separating

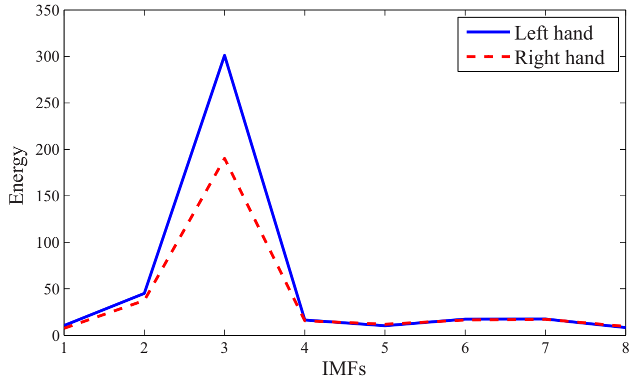

Fig. 4: Energy of all the 8 IMFs.

these, we need to perform time frequency analysis to know in which time what frequency components are more activated. For analyzing the time frequency property, well established short time Fourier transform (STFT) is used which gives the opportunity to view signal spectra at different time frames or windows.

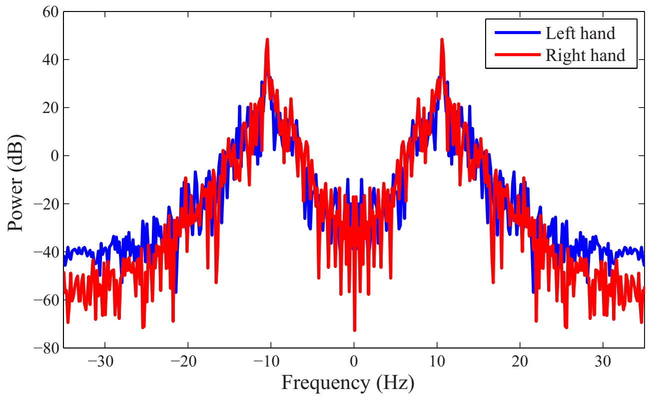

Fig. 5: PSD of third IMF for left and right hand movements.

Short time Fourier transform is a process to observe how the frequency content of a signal changes over time [23]. It is calculated by dividing the long signal into shorter time segments and then computing the spectrum of each segments or frames. If the signal $x(t)$ is pre-windowed around a time instant t and the Fourier transform is calculated at each time instant t, then the STFT of the signal $x(t)$ is expressed as,

$$STFT(t,f) = \int_{-\infty}^{+\infty} x(\tau)h(\tau - t)e^{-if\tau}\, d\tau \tag{2}$$

where $h(t)$ is the window function [24].

In this work, the STFT is applied to the 3rd IMF for 8 frames where the Fourier transform is calculated in each frame of the IMF. Let "$S_N(c)$" denotes the STFT outputs where N indicates the frame number and c is the channel index (i.e., $C3$ or $C4$). For example, $S_1(C4)$ indicates first frame of the STFT for $C4$ channel. Fig. 6 (a) represent the magnitude spectrum of $S_2(C3)$ of the third IMF. From the figure it is easily visible that the magnitude spectrum of $S_2(C3)$ of the third IMF is clearly distinguishable for left and right hand imagery movements and in particular, the peak value for left hand movement is much higher than for right hand imagery movement.

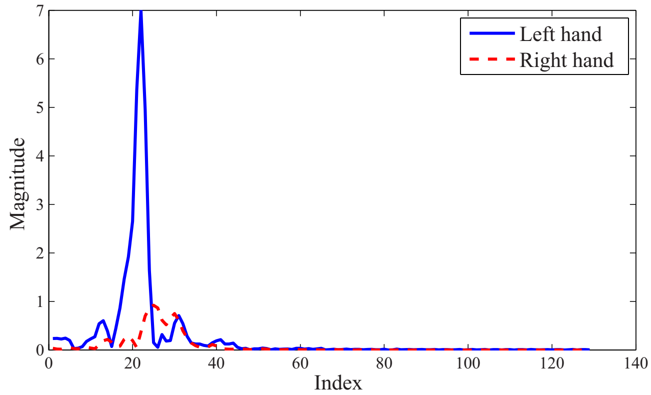

Fig. 6: Magnitude spectrum of STFT (2^{nd} frame).

Along with peak, entropy has been used as the second feature which is a measure of randomness. Entropy is a common concept in signal processing which can provide useful information about the underlying dynamical process associated with the signal [25]. In our study most common Shannon entropy has been used.

The Shannon entropy is defined as [26]

$$E(S) = -\sum_i S_i^2 \log(S_i^2) \tag{3}$$

where S_i are the coefficients of signal S in an orthonormal basis.

However, in our study it has been found that the peak and entropy of second, third and fifth frames varies significantly for left and right hand movements. In addition to that, instead of using individual peaks and Shannon entropy, we have defined a combined feature space as the product of Shannon entropy and peak value. This new feature space has two advantages. One is the separability of this combined feature space between two classes is better than the individuals (very small p-values of Kruskal-Wallis test ensure this). The second advantage is that the length of feature vector is now reduced to half compared to separately using entropy and peak and thus keeping the feature dimension smaller.

If FT indicates the feature vector, then it can be expressed as

$$FT = [F_N(C3), F_N(C4)]^T \tag{4}$$

where $F_N(C3)$ and $F_N(C4)$ for $C3$ and $C4$ channels can be expressed as

$$\begin{aligned} F_N(C3) &= max\left(abs\left(S_N(C3)\right)\right) \times E\left(abs\left(S_N(C3)\right)\right) \\ F_N(C4) &= max\left(abs\left(S_N(C4)\right)\right) \times E\left(abs\left(S_N(C4)\right)\right) \end{aligned} \tag{5}$$

here, max indicates the maximum value, abs signifies the absolute value of $S_N(c)$, 'E' is the entropy defined in (3) and $N = 2, 3, 5$.

The feature quality of FT has been statistically justified using Kruskal-Wallis p-values. The p-values of Kruskal-Wallis hypothesis testing for left and right hand imagery movements on train data set have been given in Table I. The hypothesis about the p-values is that the value $p<0.05$ indicates that at least one sample mean is significantly different than the other sample means statistically [27]. From Table I, it is clear that the p-values are very small. Hence it can be concluded that the features have good distinguishable property statistically.

TABLE I: Results of Kruskal-Wallis test

Features	P-value
$FT(1)$	2.1285e-08
$FT(2)$	4.3928e-07
$FT(3)$	2.2209e-04
$FT(4)$	4.7928e-07
$FT(5)$	2.3370e-06
$FT(6)$	0.0387

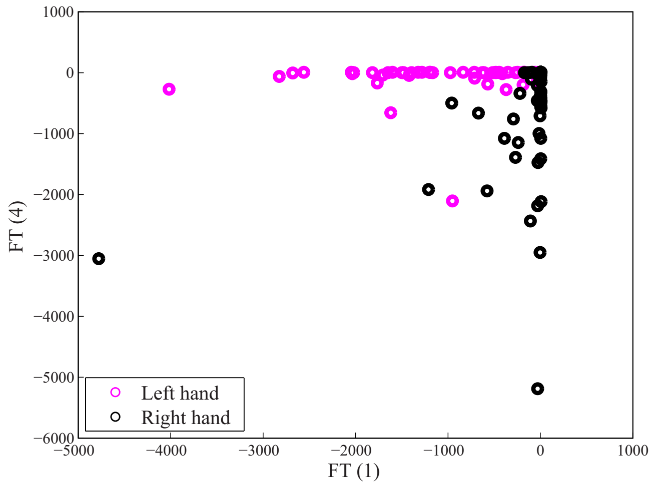

Fig. 7: Scatter plots for FT(1) and FT(4) for two imagery movements.

Apart from Kruskal-Wallis [28] p-values, scatter plots, box plots and histograms are provided to further illustrate the classification property of the extracted features. Fig. 7 represents the scatter plots of $FT(1)$ and $FT(4)$, where $FT(1)$ is the first element of the feature vector FT. In Fig. 7, magenta and black circles indicate feature values during left and right hand imagery movements, respectively which have significantly different values during different imagery hand movements and as a result, they can be regarded as good separable features to be used in classifiers.

Fig. 8 shows the box plots of $FT(1)$ and $FT(2)$, respectively for left and right hand imagery movements on the train data set. The box plots have non overlapping notches which indicates that the features have distinct values for the two specific movements.

Fig. 9 presents the histograms of $FT(1)$ and $FT(5)$ during left and right hand imagery movements where dark brown and blue color indicate left and right hand, respectively. From the histograms, it can be concluded said that the corresponding histograms span different regions which is the evidence that the features show different values during two specified motor

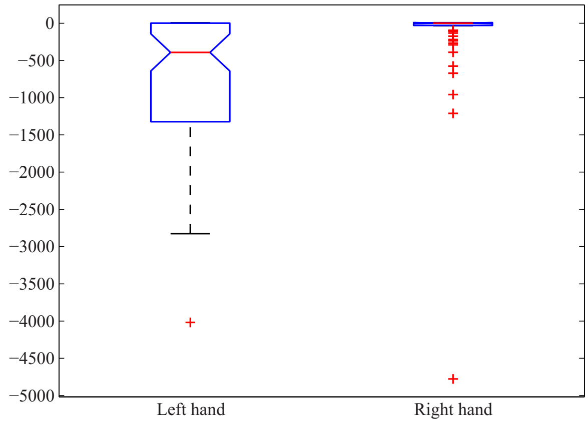

Fig. 8: Box plots of F(C3) for left and right hand.

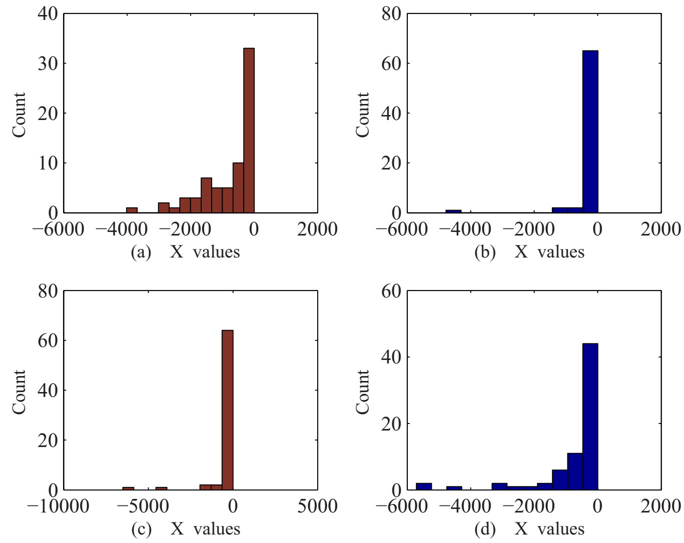

Fig. 9: (a)-(b) Histograms of $FT(1)$ for left and right hand imagery movements, respectively; (c)-(d) histograms of $FT(2)$ for left and right hand imagery movements, respectively.

imagery movements. For example, $FT(1)$ has peak around 0 for right hand imagery movements while for left hand imagery movements, it spans the region between -2000 to 0.

The scatter plots, box plots, histograms and the p-values of Kruskal-Wallis analysis justify that the extracted features (i.e., elements of FT) have significantly distinguishable values for left and right hand motor imagery movements. In other words, the features have good between-class distance and small within-class variance in the feature vector space [29] and as a result, they can be used as suitable features to classify two hand movements.

B. Classification Using the kNN Classifier

For any classification problem, there are two main parts - first is feature extraction and second is classification. If suitable features can be extracted, then a simple classifier can provide the desired outcome. Among different classifiers, kNN classifier performs best in our study. Among the various methods of supervised statistical pattern recognition, the Nearest

Neighbor rule achieves consistently high performance, without a priori assumptions about the distributions from which the training examples are drawn [30]. In order to classify a sample trial vector X which has unknown class, kNN classifier ranks the sample trial's neighbors among the training trial vectors and uses the class labels of the K most similar neighbors to predict the class of the new test trial [31]. The classes of these neighbors are then weighted according to the similarity of each neighbor where the similarity index is the cosine value between two document vectors of Euclidean distance. The cosine similarity index is defined as

$$sim(X, D_j) = \frac{\sum_{t_i \in (X \cap D_j)} x_i \times d_{ij}}{||X||_2 \times ||D_j||_2} \tag{6}$$

where X is the test or unknown trial; D_j is the j-th training trial; t_i is shared by both X and D_j; d_{ij} is the weight for t_i in training sample D_j whereas x_i is the weight for t_i in X. The l_2 norm of X is defined as

$$||X||_2 = \sqrt{x_1^2 + x_2^2 + x_3^2 + ...} \tag{7}$$

V. Experimental Analysis

Since the cue was given at t= 3 sec and lasted for 6 seconds, so each epoch for a single channel has a duration of 6 seconds with $6 \times 128 = 768$ data points since the sampling frequency is 128 Hz. We have used 3 features from each channel. So the number of feature element is six per epoch and the train and test feature matrix has dimension of 140×6 which is fed to kNN classifier. kNN classifier has been trained with train data set and after training, the label of test data set has been determined. These predicted test labels are compared with ground truth provided by BCI 2003 organizer. The experiments are carried out using MATLAB 2013b [27] on Windows-7 32 bit platform having 1 GB RAM and 2.93 GHz Intel Core 2 Duo processor.

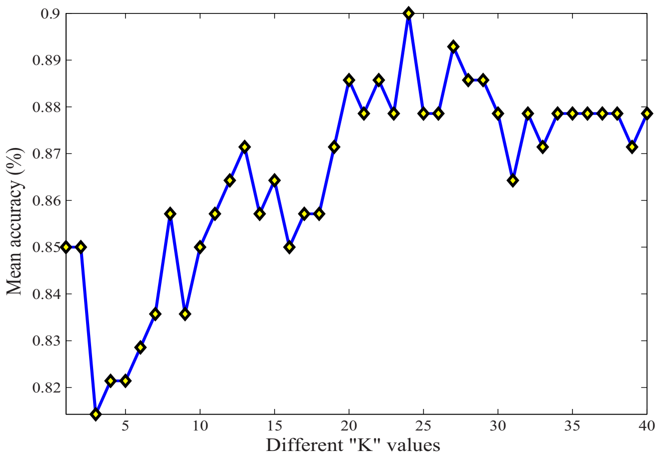

Fig. 10: Mean accuracy(%) vs different "K" values.

kNN classifier has two parameters to tune: the distance parameter and the "K" value. The distance parameter is selected as "cosine" which means one minus the cosine of the included angle between points [27]. For choosing the optimum "K" value, we have varied the value of "K" and it is noted that for our classification problem, K= 24 gives the highest accuracy of 90.00%. Fig. 10 shows the variation of "Mean accuracy (%)" for different "K" values.

TABLE II: Performance Evaluation of Different Classifiers

Classifier Name	Parameter	Mean Accuracy (%)
Discriminant Analysis	Linear Distance	85.71
	Diaglinear Distance	78.57
	Quadratic Distance	79.29
	Diagquadratic Distance	75.00
	Mahalanobis Distance	81.43
Naive Bayes	Normal Distribution	75.00
SVM	Linear Kernel	85.71
	Radial Basis	89.29
kNN	Euclidean Distance	82.86
	Cityblock Distance	83.57
	Cosine Distance	90.00

In this paper, we have also conducted performance comparison of different classifiers using our method. The classifiers considered includes is includes support vector machine (SVM), discriminant analysis (DA), Naive Bayes and k-Nearest Neighbor (kNN) with various parameters. Table II provides the outcomes of different classifiers. It is seen, kNN classifier with "cosine" distance gives better mean accuracy than other classifiers with varying parameters.

TABLE III: Comparison of mean accuracy of different methods.

Method	Proposed by	Classifier	Mean Accuracy (%)
PSD	Solhjoo et al. [10]	Mahalanobis distance	63.1
		Gaussian classifiers	65.4
		LDA	65.6
Raw EEG	Solhjoo et al. [11]	HMM	77.50
AAR	Tavakolian et al. [12]	Bayes Quadratic	82.86
		BGN	83.57
		MLP	84.29
Morlet wavelet	Lemm et al. [13]	Bayes Quadratic	89.29
Proposed method (MEMD + STFT)		kNN	90.00

Finally, Table III compares the mean accuracy of the proposed method with those of the several other methods. It is observed that using simple kNN classifier, the proposed method provides better mean accuracy in detecting left and right hand motor imagery movements than others.

VI. CONCLUSIONS

In this paper a hybrid method has been proposed to distinguish the left and right hand imagery movements which offers a promising support for an important application in BCI. EEG signals have been successfully classified by extracting suitable features after employing multivariate empirical mode decomposition followed by short time Fourier transform. Graphical and statistical justification of using the extracted features has been provided with a number of scatter plots, box plots, histograms and p-values of Kruskal-Wallis test. Among various types of classifiers like SVM, kNN, Bayes classifier and LDA, kNN provides the highest accuracy of 90.00% on test data set. Finally the performance has been compared with several other recent methods available in EEG literature and shown to be superior than others.

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